

# Natural Language Technology

Introduction

# About Me – Robert Ross

*aka, Rob.*

*Not, Sir, Professor, or any other honorific*

Senior Lecturer – School of Computer Science

Lead – Cognitive Systems Research Group

Active National Funding (Sep 2026)

- Research Ireland ADAPT Research Centre and Rinn AI
- Research Ireland Spoke on Motor Neuron Disease
- Deputy Director ARC Hub for ICT
- Enterprise Ireland Funded Innovation Partnerships and Com Funds

Research Areas:

- Situated Interactive Systems (Talking Robots / Games Characters)
- Applied Computer Vision Applications



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# NLP always in the news!



## AI startup Anthropic agrees to pay \$1.5bn to settle book piracy lawsuit

Settlement could be pivotal after authors claimed company took pirated copies of their work to train chatbots



- Jumping In
- Course Scope
- Prerequisites
- Course Structure
- Assignments and Assessment
- Coding
- Lab: Model Training with Collab

AGENDA

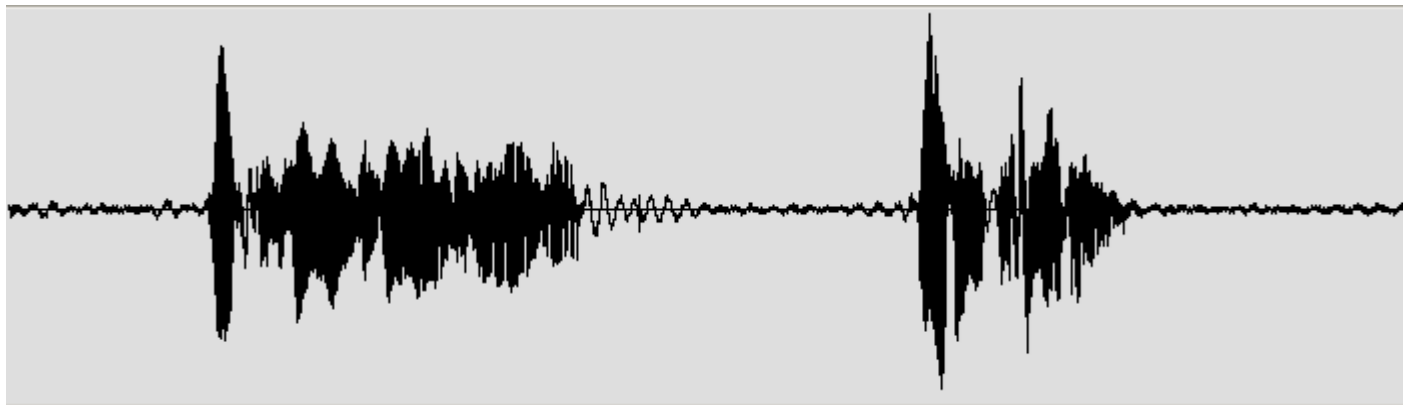


Course Scope



# Course Scope

- Goal: Study key concepts in Natural Language Technology
  - Emphasis on practice
  - Linking out to theory
  - Linking out to related modules



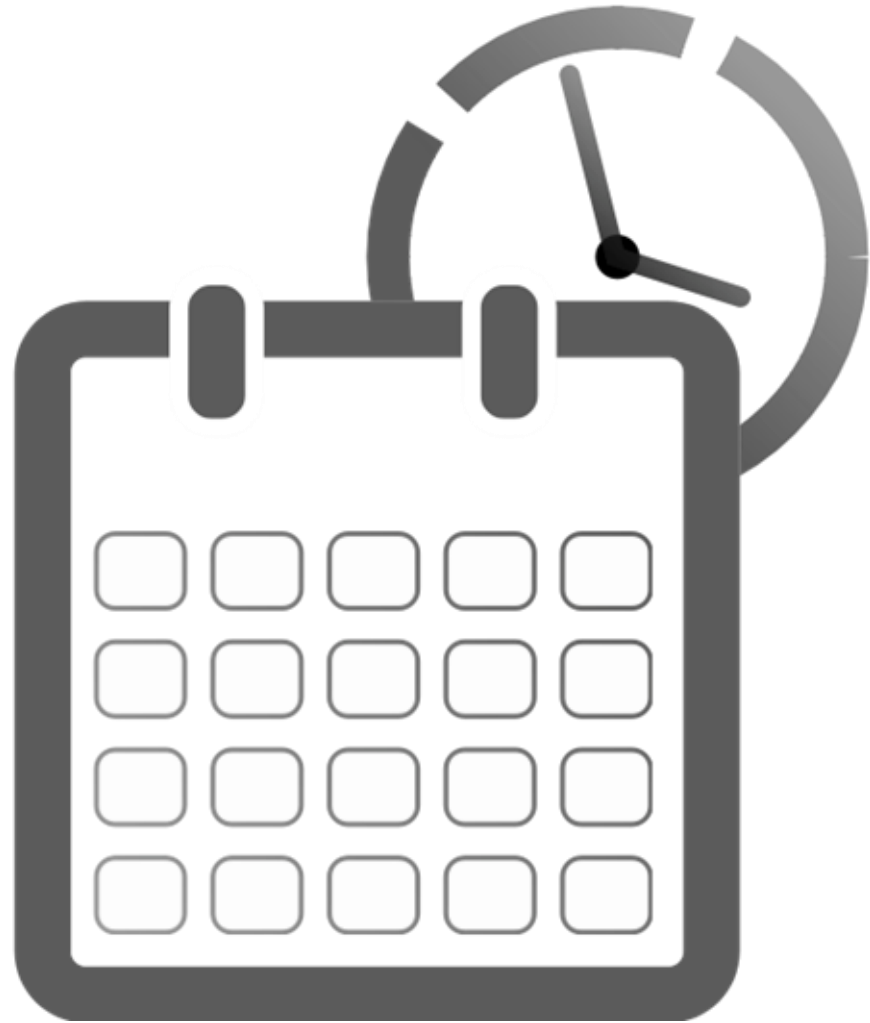
# Course Scope

- Goal: Study key concepts in Natural Language Technology
  - Emphasis on balanced theory and practice
- Lecture Notes: Mostly Slides
- Not Covered
  - Digital Signal Processing (see Sean O'Leary)
  - Computational Linguistics (the old way – lots of linguistics)
  - Machine Learning and Deep Learning (see other modules)
- Learning Resources
  - Course is mostly self contained
  - Links to external resources will be given
  - Use of 3rd party services will be a part of the course

# Prerequisites

- Good programming experience
- All assignments to be coded in Python
- Machine Learning or AI is not a prerequisite
- This module is suited to both Data Analytics and Advanced Software Development students
- Laptop
  - I strongly encourage all students to set up their own local python environments to learn
- Willingness to interact
  - We don't all like groups, trust me! But they are a good way to learn.

# Course Structure



# Course Structure

- ***Week 1 - Introduction - the Challenge of Language***
- ***Week 2 – Language Input / Speech Recognition***
- ***Week 3 – Language Output / Speech and Image Production***
- ***Week 4 – Representing the Meaning of Text***
- ***Week 5 – Processing Text***
- ***Week 6 – Information Retrieval***
- ***Week 7 - Review Week Bank Holiday***
- ***Week 8 – Language Modelling***
- ***Week 9 – Large Language Models***
- ***Week 10 – Prompts and RAG***
- ***Week 11 – Agentic AI and Orchestration***
- ***Week 12 – Bias and Ethics***
- ***Week 13 – Review Week***

Provisional and will  
change!!!!

# Delivery

- 5 ECTS
- Three hours contact per week
- 3 Hours on Monday Night
  - 2 Lecture
  - 1 Lab
- Assignments
  - Examination: 50%
  - CA: 50%



# Coding

- Python plus requested libraries and frameworks
  - Python 3.12+
  - Use Virtual Environments
- Server Environment
  - Most code can be run locally
  - Google Collab can generally be used for most experimental work

# On the Assignment

- Initial Plan Design
- In Class Lab Review
- Github and Shared Report
- Final Oral Defense that caps other elements

| Component                                  | %  | Format                                                                   |
|--------------------------------------------|----|--------------------------------------------------------------------------|
| Initial Check in and Experimental Plan     | 5  | Initial commit and experimental plan, w/c 10 <sup>th</sup> Nov           |
| Online checkpoint                          | 5  | Individual 5-minute conversation online, w/c 24 November approx          |
| Project — Parts 1 to 4, submitted together | 25 | Own time, one submission on 14 December approx                           |
| Oral assessment                            | 15 | Individual 20-minute conversation in person, Week of 12 January approx.. |
| Assignment total                           | 50 |                                                                          |

See Content Deck

# Labs for Week 1

## Part 1:

Download the notebook for week 1.

Upload it onto google Colab and execute content

Switch Runtime model on Google Colab between GPU and CPU.

## Part 2:

Install Python 3.12 on your own machine

Make sure Python is being added to your PATH

Run python from the command line and verify the version

